

ERP-PSM System Documentation

Complete Architecture, Technology Stack, Module Workflows & Developer Extension Manual

APPLICATION	ARCHITECTURE	DOMAIN FOCUS	COMPLIANCE
ERP-PSM Suite	PHP CI3 MVC & MariaDB	Tenders & Landed Cost	VAT, NBR & Double-Entry

1. Technology-Wise Architectural Breakdown

SYSTEM STACK

ERP-PSM employs a robust Model-View-Controller (MVC) decoupled design, pairing lightweight server execution with dynamic client-side DOM manipulation and an ACID-compliant double-entry ledger.

BACKEND LAYER

PHP & CodeIgniter 3

- **Controllers:** Standardized REST-like endpoints with session guarding (`zazu_logged_in`).
- **Routing:** Kebab-case SEO friendly URI aliases registered in `application/config/routes.php`.
- **Security:** Granular RBAC permissions, CSRF tokens, and query parameter escaping via `escape_str()`.
- **Libraries:** Composer integration with `firebase/php-jwt` for secure web tokens.

DATA LAYER

MySQL / MariaDB (InnoDB)

- **Header-Line Relationships:** Master transaction tables linked with line item child tables.
- **Double-Entry Ledger:** Chart of accounts (`cb_account_head_info`) with auto-balancing debits & credits.
- **Soft-Deletes:** Standardized `status = 'Active' / 'Deleted'` preserving full audit trails.
- **Foreign Key Integrity:** Cascading joins preventing orphan financial entries.

PRESENTATION LAYER

Bootstrap 3 & AdminLTE

- **Dynamic DOM:** jQuery & AJAX dynamic calculations (unit rates, margins, freight, VAT).
- **DataTables Suite:** Server-side & client-side filtering, sorting, and Excel/PDF export.
- **Modular Views:** Split into layout templates (`page/`) and dedicated scripts (`inc-js/`).
- **Components:** Select2 searchable dropdowns, datepickers, and modal transaction entry forms.

STORAGE & PRINT

Documents & A4 Output

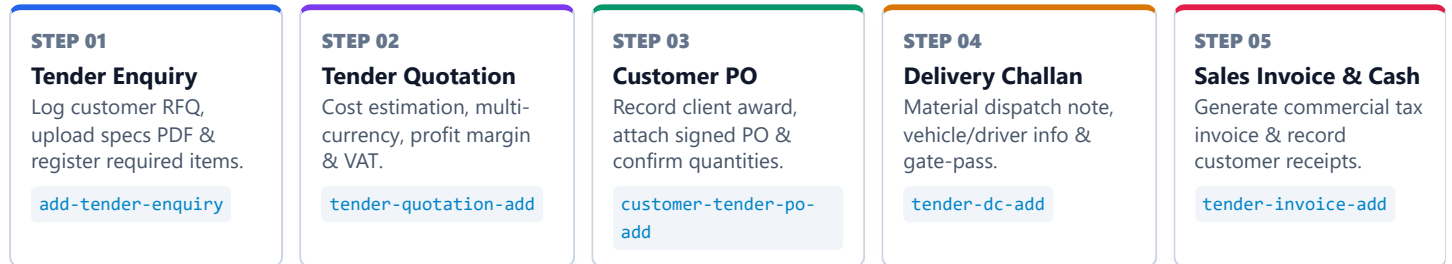
- **File Vault:** Isolated sub-folders for `tender-documents/`, `vendor-quotations/`, and `uploads/`.
- **Print Engine:** Pixel-perfect A4 portrait CSS rules with page-break controls for all official notes.
- **Vouchers:** Formatted print sheets for Delivery Challans, Tax Invoices, Receipts, and POs.
- **Web Server:** Apache 2.4+ with clean URL rewrites through `root .htaccess`.

2. Dual Synchronized Operational Workflows

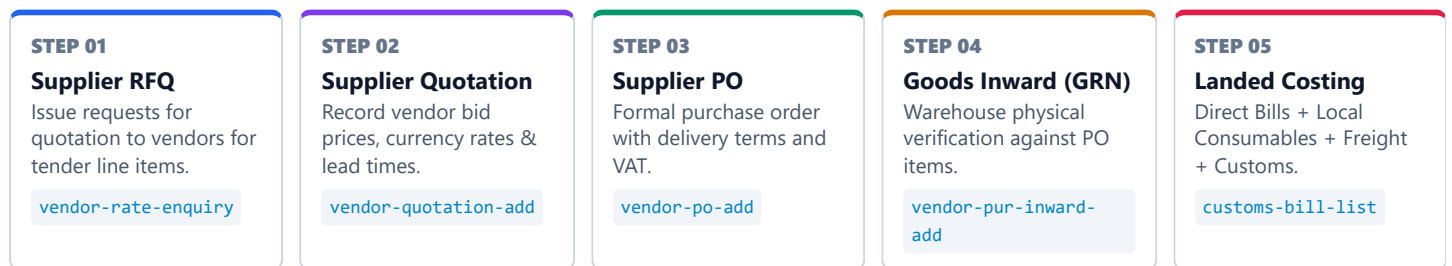
BUSINESS LIFECYCLES

The system synchronizes two parallel operational tracks: **Customer Tender-to-Cash** and **Supplier Procurement with Landed Costing**, tying them together at the project reporting layer.

A. Customer Tender-to-Cash Cycle



B. Supplier Procurement & Complete Landed Costing



3. Core Module & Functional Matrix

MODULE REFERENCE

MODULE	KEY ROUTES & ACTIONS	FUNCTIONAL CAPABILITIES	ACCOUNTING LINK
Dynamic RBAC	/menu-management /role-permission	Hierarchical menu tree builder, custom user roles, granular view/add/edit/delete access control.	SYSTEM SECURITY
Master Setup	/company-list /customer-list /items-list	Multi-bank setup (IBAN/Swift), tax registration, vendor/customer opening balances, item catalog, HSN & UOM.	MASTER LEDGERS
Customer Tenders	/tender-enquiry-list /tender-quotation-add /tender-invoice-add	Tender bidding, margin markup, PO conversion, vehicle dispatch note (DC), commercial VAT invoice & payment receipts.	ACCOUNTS RECV.
Procurement & Landed Cost	/vendor-po-add /vendor-pur-inward-add /customs-bill-list	Supplier PO, Goods Receipt Note (GRN), local consumable bills, delivery partner shipping fees, and customs duty.	ACCOUNTS PAY.
Accounts & Treasury	/inward-list /outward-list /contra-entry /petty-cash	Non-sales receipts, administrative expenses, internal bank-to-bank transfers, petty cash vouchers, credit & debit notes.	GENERAL LEDGER
Reports & Intelligence	/tender-enquiry-summary-report /customer-statement-report /sales-nbr-report	Full Project Profitability Matrix (Invoiced Sales vs. Landed Costs), Statements of Account (SOA), NBR & VAT compliance.	AUDIT & TAX

4. Developer Guide: Step-by-Step Feature Implementation

DEVELOPER MANUAL

To extend ERP-PSM with new features, modules, or transaction books, follow this standardized 8-step pipeline to ensure full compliance with the existing architecture and security models.

1 Database Table Design

Follow the standard naming pattern (<entity>_info) with auto-increment ID, company foreign key, soft-delete status, and timestamps:

```
CREATE TABLE `sample_feature_info` (  
  `sample_id` INT(11) NOT NULL AUTO_INCREMENT,  
  `company_id` INT(11) NOT NULL,  
  `title` VARCHAR(255) NOT NULL,  
  `amount` DECIMAL(15,2) DEFAULT '0.00',  
  `status` ENUM('Active','Inactive','Deleted') DEFAULT 'Active',  
  `created_by` INT(11) DEFAULT NULL,  
  `created_at` DATETIME DEFAULT CURRENT_TIMESTAMP,  
  `updated_at` DATETIME DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP,  
  PRIMARY KEY (`sample_id`),  
  KEY `idx_company` (`company_id`),  
  KEY `idx_status` (`status`)  
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4;
```

2 Register Routes in application/config/routes.php

Map clean URL endpoints to your controller methods:

```
$route['sample-feature-list']           = 'master/sample_feature_list';  
$route['sample-feature-list/(:num)']    = 'master/sample_feature_list/$1';  
$route['sample-feature-save']           = 'master/sample_feature_save';  
$route['sample-feature-delete/(:num)']  = 'master/sample_feature_delete/$1';
```

3 Controller Action in application/controllers/Master.php

Enforce session authentication, query active records, and load the 5-layer standard view stack:

```
public function sample_feature_list($id = null)  
{  
    if (!$this->session->userdata('zazu_logged_in')) {  
        redirect('login');  
    }  
  
    $data['title'] = "Sample Feature Management";  
    $data['edit_id'] = $id;  
  
    // Fetch active records  
    $data['records'] = $this->db->get_where('sample_feature_info', ['status' => 'Active'])->result_array();  
  
    // Standard 5-tier view assembly  
    $this->load->view('inc/header', $data);  
    $this->load->view('inc/admin-menu');  
    $this->load->view('page/master/sample-feature-list', $data);  
    $this->load->view('inc/footer');  
    $this->load->view('inc/inc-js/master/sample-feature-list.inc');  
}
```

4 View Template in application/views/page/master/sample-feature-list.php

Construct the AdminLTE container box with responsive DataTable layout and modal forms:

```

<div class="content-wrapper">
  <section class="content-header">
    <h1>Sample Feature <small>Management</small></h1>
  </section>
  <section class="content">
    <div class="box box-primary">
      <div class="box-body">
        <table id="sampleTable" class="table table-bordered table-striped">
          <thead>
            <tr><th>#</th><th>Title</th><th>Amount</th><th>Actions</th></tr>
          </thead>
          <tbody>
            <!-- Row records or dynamic AJAX populate -->
          </tbody>
        </table>
      </div>
    </div>
  </section>
</div>

```

5 Client-Side Script in application/views/inc/inc-js/master/sample-feature-list.inc

Initialize DataTable and bind AJAX submission handlers:

```

<script>
$(document).ready(function() {
  $('#sampleTable').DataTable({ responsive: true, pageLength: 25 });

  $('#btnSaveSample').on('click', function(e) {
    e.preventDefault();
    $.ajax({
      url: "<?= base_url('sample-feature-save'); ?>",
      type: "POST",
      data: $('#sampleForm').serialize(),
      dataType: "json",
      success: function(res) {
        if(res.status === 'success') { location.reload(); }
        else { alert(res.message); }
      }
    });
  });
});
</script>

```

6 Menu & Permission Setup via UI

No hardcoding needed in sidebar templates!

1. Go to **Menu Management** (/menu-management), add "Sample Feature" with route sample-feature-list.
2. Go to **Role Permissions** (/role-permission), check the access checkboxes for the desired user roles and hit save.